Bukang-liwayway v3.2 — Printing & Assembly Guide

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Before you start

Bukang-liwayway ("Sunrise") is Sol Prosthetics' Model 1: a wrist-driven hand for children about 9 and under. Bending the wrist down pulls four braided lines and the fingers close; bending it back lets elastic cords open them. The thumb sits on a friction hinge you set by hand. The palm side closes with a soft TPU cover, and the child's hand slides in through the open wrist end.

- **This is a test print.** v3.2 is the second test version, with the fixes from the first full print (v3.1g). It has not been fitted to a child yet. Treat it as a trial and check everything in the Tuning section.
- **Print at 100% only.** Never scale the files in the slicer: the screw holes, insert holes and joint gaps would change size. Ready-made size sets, each with its own hardware card, come after the test fit.
- **It fits** a child's hand about 55 mm across the knuckles and 129 mm from wrist crease to middle fingertip, with a palm up to about 49 mm long from the wrist crease. Wrist about 97 mm round, forearm about 110 mm.
- **Materials:** PETG for every rigid part (never PLA, it softens in the heat) and TPU 95A for the palm cover.
- If you can, fit the first hands with a physical therapist or prosthetist. Every residual limb is different.

Parts to print

Print one of each part below from the `stl` folder (GitHub or the website). Every file is already turned to its print direction, so load it as it is and don't rotate it. Print the two `_print` versions: they have snip-off supports built in.

File	Part	Material	Size (mm)	Supports
`palm_back_print.stl`	Back of the hand, with a snip-off rim support	PETG	74 × 52 × 77	Normal + its blocker; brim

File	Part	Material	Size (mm)	Supports
`palm_flap_print.stl`	Palm cover, with snip-off fins and sole bar	**TPU 95A**	71 × 26 × 51	Build plate only; brim
`cuff.stl`	Forearm cuff	PETG	61 × 39 × 109	A few
`thumb.stl`	Thumb	PETG	24 × 26 × 30	Build plate only; brim
`finger-index-1/2/3.stl`	Index finger, 3 segments	PETG	up to 18 × 30 × 30	Build plate only; brim
`finger-middle-1/2/3.stl`	Middle finger, 3 segments	PETG	up to 17 × 34 × 35	Build plate only; brim
`finger-ring-1/2/3.stl`	Ring finger, 3 segments	PETG	up to 15 × 31 × 32	Build plate only; brim
`finger-little-1/2.stl`	Little finger, 2 segments	PETG	up to 14 × 25 × 24	Build plate only; brim

`palm_back.stl` and `palm_flap.stl` are the same parts without the print aids, for reference. `HARDWARE.md` in the same folder lists every screw and insert for this size.

Segment 1 is the one that joins the palm; segment 3 (2 on the little finger) is the fingertip with the nail. Plastic needed: roughly 100–135 g of PETG plus supports, and under 10 g of TPU.

Print settings

A 0.4 mm nozzle and 0.2 mm layers are assumed for every part.

Setting	PETG parts	TPU palm cover
Walls	5	4, so the 3 mm snap pegs print solid
Top / bottom layers	Your usual	4 each
Infill	40–50% gyroid (fingers, thumb, back of hand); 30% (cuff)	30% gyroid (the 2 mm cover is almost all walls and top/bottom layers)
Speed	Normal for your PETG	Slow, 20–30 mm/s

Setting	PETG parts	TPU palm cover
Retraction	Normal	Little or none
Surface	Normal	**Fuzzy skin** on the outer grip face only, so the palm isn't slippery. Never on the snap pegs: fuzzy skin adds about 0.3 mm and the 3.0 mm pegs then jam or tear in their 3.2 mm holes. Paint it on the grip face, or use a modifier to turn it off round the pegs. If that's a bother, skip it; TPU 95A is grippy anyway.
Supports	Back of the hand: normal supports with its blocker loaded (below). Fingers and thumb: on the build plate only. Cuff: a few	Under the snap pegs
Brim	5 mm: back of the hand, fingers and thumb	5 mm

Per-part tips:

- **Screw holes never need support.** Every screw hole prints lying sideways and is small (2.6–3.4 mm, with 6.2 mm head recesses), so each layer only bridges a few millimetres over the top of the hole.
- **Fingers and thumb:** they stand at 45°, knuckle end down and nail side up, so no support scars land on the faces you see. Supports on the build plate only: that removes every support the slicer wants inside the screw holes and keeps the ones under the low knuckle end. Use a brim, because each part touches the bed only a little.
- **Back of the hand:** it stands on its wrist end, so the line tunnels run almost straight up. It touches the bed only under the two wrist ears, and the rim across the back floats about 8 mm up, right where the line tunnels come out. `palm_back_print.stl` holds that rim on 14 small posts with thin necks, all on solid plastic between the tunnel exits, standing on a thin sole that also keeps the tall part from tipping over. Keep normal supports on (the inside of the knuckles and the underside of the thumb mount need them) and load `palm_back_print_support_blocker.stl` with it, so no support goes into a hole or a tunnel. Supports never touch the logo.
  - **Creality Print / OrcaSlicer / PrusaSlicer:** drag both files in together and answer **Yes** to loading them as one object with multiple parts. Right-click the blocker part → **Change type → Support blocker**.

    - **Cura:** turn off "Automatically drop models to the build plate" and "Ensure models are kept apart" before loading, so both keep their positions. Select the blocker → **Per Model Settings** → **Don't support overlaps**.
- **Palm cover:** print `palm_flap_print.stl`. Dry the TPU first if it has been sitting out; wet TPU strings and blobs.
- **All parts:** every file is already a clean, closed solid. Don't run the slicer's mesh-repair, "fill holes" or "remove small features" options on it.

Hardware and materials

All screws are stainless M3 from a hardware store, and no nuts are needed. Most screws thread into a 2.6 mm pilot hole in the plastic; the knuckle axle and the three cover screws thread into brass heat-set inserts. This is the list for the 100% size; `HARDWARE.md` in the `stl` folder has the list for the size you print.

Item	Qty	Where it goes
M3 × 45 **button-head** screw (ISO 7380)	1	Knuckle axle: in from the index side, through the index, middle and ring knuckles
M3 × 12 button-head screw	6	Middle and tip joints of the index, middle and ring fingers
M3 × 10 button-head screw	2	Little finger: knuckle and tip joint
M3 × 20 button-head screw	1	Thumb hinge (threads straight into the plastic, no insert)
M3 × 6 button-head screw	2	Wrist hinge, one through each cuff arm
M3 × 4 **countersunk (flat-head)** screw	4	Line clamps on top of the cuff
M3 × 5 **countersunk (flat-head)** screw	3	The palm cover's little-finger edge
M3 brass **heat-set insert**, 4.2 mm across × 4 mm long (double-knurled)	4	Index knuckle inner arm (faces the middle finger); 3 in the cover step
PE braided fishing line #4, 50 lb (about 0.32 mm)	about 2 m	4 flexor lines (close the fingers)
1 mm clear round elastic cord	about 1.5 m	4 return cords (open the fingers)

Item	Qty	Where it goes
Bicycle inner tube, about 1 mm (butyl)	1 small piece	5 grip pads and 2 thumb-hinge washers
E6000 or contact cement	a little	Gluing the grip pads
Sew-on hook-and-loop strap, 25 mm	2	Cuff straps
Washable foam or mesh liner, 1–1.5 mm	1 piece	Inside the cuff and the back of the hand

Tools you need

- 2 mm hex key (fits every button-head and flat-head screw), with a long one or a hex bit for the 45 mm knuckle axle
- Soldering iron with a heat-set insert tip (or a plain tip), set to about 240 °C for PETG
- 3.5 mm drill bit, turned by hand (a pin vise or just your fingers), for cleaning the 3.4 mm screw holes
- Flush cutters and a craft knife for removing supports, the snip-off necks and brims
- Fine sandpaper or a needle file
- Thin wire or a long darning needle for threading lines through the tunnels
- Lighter or a drop of super glue for sealing knots in the braided line
- Scissors, rubbing alcohol and a cloth
- Needle and strong thread for sewing on the straps

After printing

- [ ] **Back of the hand:** snip each of the 14 necks flush at the rim with flush cutters, lift the support and sole off, and sand the marks smooth. Check that every line-tunnel exit on the rim is clear.
- [ ] **Palm cover:** snip the small tabs off the fins and the sole bar, and shave any nub flush.
- [ ] Remove the other supports and brims. Pay attention inside the knuckle cups, the thumb mount slot and under the cover's snap pegs.
- [ ] Sand smooth any edge that will touch skin: the open wrist end of the back, the cuff ends and strap slots.
- [ ] **Clean the screw holes.** The top of a hole printed without support can sag 0.1–0.2 mm. By hand, run the 3.5 mm drill bit through the 3.4 mm holes the screws slide through (finger joints, knuckle axle, thumb). **Don't drill the 2.6 mm pilot holes** the screws

thread into: that would loosen the grip. Then run an M3 screw through every pilot hole once to cut the thread, and back it out.

- [ ] Push a piece of line through every channel and tunnel to check it is clear: the upper and lower channel in every finger segment, 8 tunnels in the back of the hand, 4 in the cuff.
- [ ] Check that every snap hole in the back's step is clear, with its small round recess on the inside.
- [ ] Dry-fit the cover in its step: it should lie flush all round with no gaps.

Assembly

Work in this order; each step leaves the next screw reachable. Tighten joints only until they move freely without wobble.

1. **Set the inserts.** Soldering iron at about 240 °C for PETG (350 °C is too hot and melts the hole oversize). Let the iron's weight push each insert in, then press it flush with something flat and cool. One goes in the index knuckle segment's inner arm (the side that faces the middle finger); three go in the cover step along the little-finger edge of the back. The thumb has **no** insert.
2. **Build each finger.** Join its segments in order (1 to 2, 2 to 3). Each joint is a rounded barrel in a cup held by one button-head screw. Put the screw in from the side with the round recess, so the head sits flush; it threads into the pilot hole on the far side. A flat stop on the back of each joint keeps the finger from bending backwards.
   - Index, middle and ring: M3 × 12 (middle joint), M3 × 12 (tip joint)
   - Little: M3 × 10 (tip joint)
3. **Knuckle axle.** Line up the index, middle and ring knuckle cups on their tongues on the back of the hand. Push the M3 × 45 in from the index side, through all three knuckles, and screw it into the index insert until its head sits in the recess. It is a plain pin through the middle and ring knuckles, so they swing freely on it.
4. **Little finger.** Slide its cup over its knuckle tongue and screw in its own M3 × 10 from the recessed side.
5. **Thumb.** Cut two small washers from inner tube. Put one on each side of the thumb's tongue, fit it into the slotted barrel on the thumb mount and screw in the M3 × 20: it runs about 10 mm into solid plastic, so it grips without an insert. Swing the thumb to the grip position you want, then tighten until it holds there.
6. **Return cords (open the fingers).** Knot the elastic in the knot pocket under the fingertip pad. Thread it through the upper channel of each segment and through its tunnel in the back of the hand. Knot it at the wrist end, just snug enough to hold the finger open. A surgeon's knot with a drop of glue holds best in the smooth elastic.

7. **Flexor lines (close the fingers).** Knot each braided line in the same fingertip pocket and seal the knot. Thread it through the lower channel of each segment, into the knuckle block and through its tunnel to the wrist end. A thin wire or long needle helps; the holes are 2–2.5 mm with 3 mm funnel mouths.
8. **Wrist.** Join the back of the hand to the cuff with the two M3 × 6 screws through the cuff's hinge arms into the hand's ear pads (the ear holes go right through).
9. **Tensioner.** Feed each flexor line into its hole in the front end of the cuff. It runs inside the wall, under its clamp, and comes out on the inside of the cuff behind the swelling. With the wrist straight and the fingers open, pull it snug and screw in its M3 × 4 flat-head clamp screw from the top until it sits flush; the screw pins the line. Trim the tail inside the cuff.
10. **Grip pads.** Cut inner tube to the recess sizes below. Sand the rubber and the recess dull, wipe both with alcohol, put a thin film of E6000 or contact cement on both and wait until tacky (2–5 minutes). Press the pad in, clamp it for 24 hours, and let it cure 72 hours before a child wears the hand.
11. **Palm cover.** Lay the cover's little-finger edge in its step and fit the three M3 × 5 flat-head screws from outside into the inserts until they sit flush. Don't overtighten; the TPU only needs to be held. Fold the cover shut and press firmly over each snap peg until its head pops through the hole in the step. You'll feel each one click.
12. **Straps and liner.** Sew the two straps onto the cuff through its strap slots, fit the liner inside the cuff and the back of the hand, and sand any edge that touches skin.

Grip pad (cut as an oval)	Cut size (mm)
Index fingertip	7 × 6.5
Middle fingertip	7 × 6.5
Ring fingertip	7 × 6.5
Little fingertip	7 × 9
Thumb tip	9.5 × 8

Tuning and test checks

In CAD, the thumb meets the index fingertip at about 50–55° of finger curl, and the hand swings from 55° down to 30° up with nothing rubbing, cover included. Check the real print against that.

- [ ] **Knuckle axle:** the index, middle and ring fingers should each swing freely on the long axle without dragging their neighbours. If the middle or ring binds, back the axle out a quarter turn.
- [ ] **Line tension:** bend the wrist down and watch all four fingers. If one closes late or not at all, loosen its clamp screw, pull its line a little tighter and re-tighten. Adjust until all four close together.
- [ ] **Opening:** with the wrist straight, every finger should spring fully open. If one stays bent, tighten its return cord knot at the wrist end.
- [ ] **Joint friction:** each joint should swing freely with no side wobble. Loosen or tighten its screw by a quarter turn at a time.
- [ ] **Pinch:** close the hand and check that the thumb meets the index fingertip. For an earlier, tighter pinch, swing the thumb a little further on its hinge and re-tighten.
- [ ] **Wrist range:** the hand should swing from about 55° down to 30° up without rubbing the cuff or the cover.
- [ ] **Cover:** open and close it ten times. Every snap should click in and the edge should stay flush.
- [ ] **Fit:** with the liner in and the cover open, the child's hand should go in through the palm side and the open wrist end without forcing, and nothing should press on the skin.

Putting it on and daily care

1. Peel the cover open from the thumb side; it stays screwed on along the little-finger edge.
2. Slide the child's hand in from the palm side; the wrist end is open, so the forearm passes straight through into the cuff.
3. Press the cover shut over each snap until it clicks.
4. Do up the two cuff straps, snug but not tight.

- **Check the child's skin every day,** especially in the heat. A closed shell hides red marks and pressure spots. If a mark doesn't fade within about 20 minutes, stop using the hand until it has been checked.
- **Wash** the cover, liner and shell with soap and water. Let them drain and dry; a drain hole at the front of the hand lets water out.
- **Keep it out of hot cars and direct sun** for long periods. PETG holds up far better than PLA, but heat still softens plastic.
- **Replace worn parts** by reprinting them: each finger segment, the thumb and the cover are separate files. Replace a flexor line if it frays.